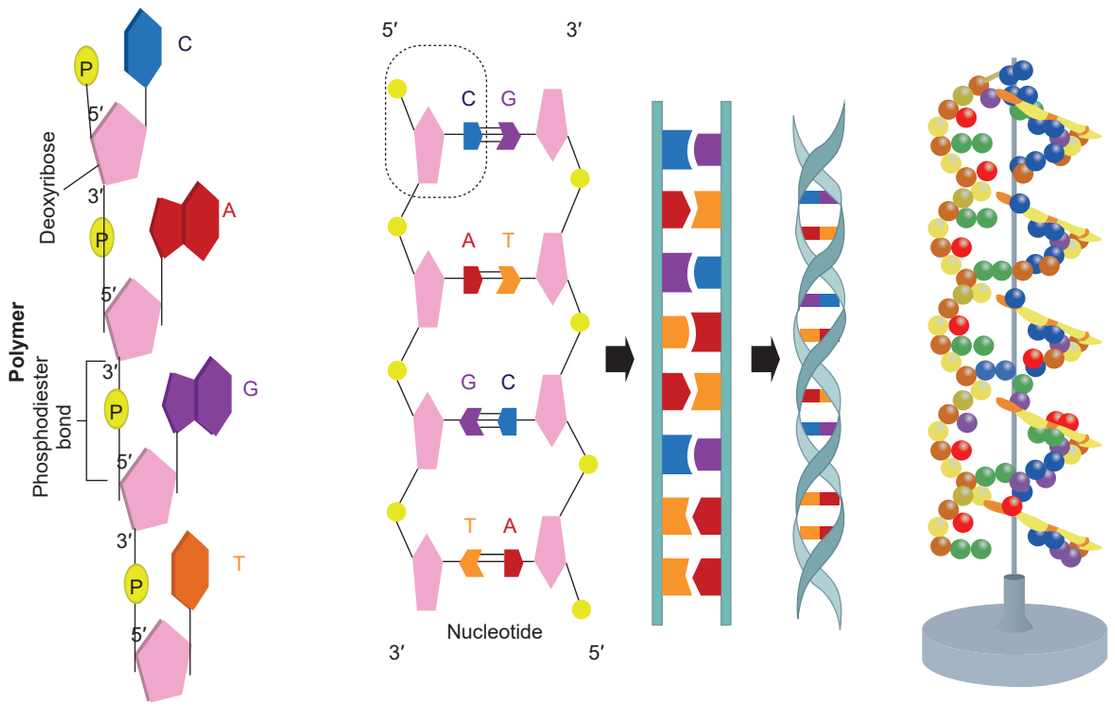


a) DNA polynucleotide chain | b) Two DNA strands bonded by hydrogen bonds | c) DNA ladder | d) DNA double helix | e) DNA around a central axis

Figure 14.7: Structure of a single polynucleotide chain of DNA, hydrogen bonding between two DNA strands, Double helix around axis

the double helix model, Nobel prize was awarded in 1962 to Watson, Crick and Wilkins. According to Watson and Crick model (Figure 14.9),

- The DNA consists of two helical polynucleotide chains wound around the same axis to form a right handed helix.
- The Purine and Pyrimidine bases of both strands are stacked inside the double helix.
- Each nitrogenous base of one strand is paired in the same plane with a base of the other strand.
- According to Watson and Crick rule Adenine base pairs with Thymine and Guanine base pairs with Cytosine.
- Two hydrogen bonds are present between A and T (symbolised as A=T)

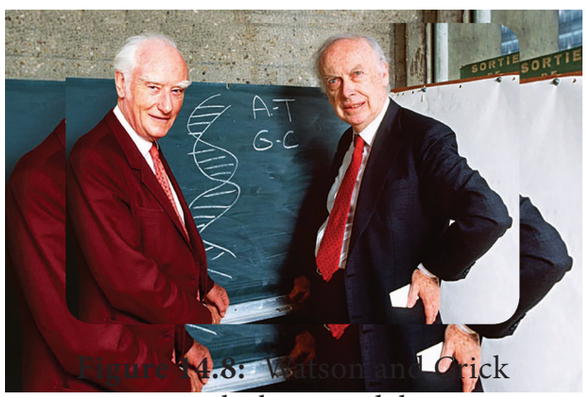


Figure 14.8: Watson and Crick with their model

and three hydrogen bonds are present between G and C (G ≡ C).

- The hydrogen bonds provide chemical stability essential to hold the two chains together.
- The specific A equal to T and G equal to C base pairing is the basis for the complementarity concept. This complementarity concept is very

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